Molecular Basis of Inheritance

At the time of Mendel the nature of the factors regulating the pattern of inheritance was not clear, and over the next hundred years the nature of the putative genetic material was investigated, culminating in the realisation that DNA is the genetic material, at least for the majority of organisms.

The two nucleic acids

- nucleic acids are polymers of nucleotides, and deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) are the two types found in living systems
- DNA acts as the genetic material in most organisms
- RNA, though it also acts as a genetic material in some viruses, mostly functions as a messenger, and has additional roles as adapter, as structural molecule and in some cases as a catalytic molecule

What this chapter covers

- the structure of DNA and its replication
- the process of making RNA from DNA, that is transcription
- the genetic code that determines the sequences of amino acids in proteins, and the process of protein synthesis, that is translation
- the elementary basis of their regulation

In this chapter

5.1 The DNA

5.7 The search for genetic material

5.10 Replication

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5.20 Regulation of gene expression

5.22 Human Genome Project

Nitrogenous base, pentose sugar and phosphate, the three parts of a nucleotide

N-glycosidic linkage joining base to sugar, making a nucleoside

Phosphoester linkage adding the phosphate, making a nucleotide

3' to 5' phosphodiester linkages joining nucleotide to nucleotide

Sugar and phosphate forming the backbone, bases projecting from it

Figure 5.A A Polynucleotide chain

5.1 The DNA

How long a DNA is

- the determination of the complete nucleotide sequence of the human genome during the last decade has set in a new era of genomics
- DNA is a long polymer of deoxyribonucleotides, and its length is usually defined as the number of nucleotides, or of base pairs, present in it, which is also a characteristic of an organism
- a bacteriophage known as ϕ ×174 has 5386 nucleotides, bacteriophage lambda has 48502 base pairs (bp), Escherichia coli has 4.6 million bp, and the haploid content of human DNA is 3.3 billion bp

5.2 Structure of a polynucleotide chain

Bases, sugar and phosphate

- a nucleotide has three components, a nitrogenous base, a pentose sugar, which is ribose in RNA and deoxyribose in DNA, and a phosphate group
- there are two types of nitrogenous base, the purines adenine and guanine, and the pyrimidines cytosine, uracil and thymine, cytosine being common to both DNA and RNA, thymine present in DNA and uracil present in RNA in place of thymine
- a nitrogenous base is linked to the OH of the 1' C of the pentose sugar through an N-glycosidic linkage to form a nucleoside, such as adenosine or deoxyadenosine, guanosine or deoxyguanosine, cytidine or deoxycytidine, and uridine or deoxythymidine
- when a phosphate group is linked to the OH of the 5' C of a nucleoside through a phosphoester linkage a corresponding nucleotide, or deoxynucleotide depending on the sugar, is formed, two nucleotides are linked through a 3'-5' phosphodiester linkage to form a dinucleotide, and more nucleotides can be joined in this manner to form a polynucleotide chain

The two ends

- a polymer so formed has at one end a free phosphate moiety at the 5' end of the sugar, referred to as the 5' end of the polynucleotide chain
- at the other end the sugar has a free OH of the 3' C group, referred to as the 3' end of the chain
- the backbone of a polynucleotide chain is formed by the sugars and phosphates, and the nitrogenous bases linked to the sugar moiety project from the backbone
- in RNA every nucleotide residue has an additional OH group at the 2' position in the ribose, and uracil is found in place of thymine, thymine being 5-methyl uracil

From Meischer to Watson and Crick

- DNA as an acidic substance present in the nucleus was first identified by Friedrich Meischer in 1869, who named it Nuclein, but due to technical limitations in isolating such a long polymer intact the elucidation of its structure remained elusive for a very long period
- it was only in 1953 that James Watson and Francis Crick, based on the X-ray diffraction data produced by Maurice Wilkins and Rosalind Franklin, proposed a very simple but famous double helix model for the structure of DNA
- one of the hallmarks of their proposition was base pairing between the two strands of polynucleotide chains, and it was also based on the observation of Erwin Chargaff that for a double stranded DNA the ratios between adenine and thymine and between guanine and cytosine are constant and equal one

5.3 The double helix

Complementarity, and what follows from it

- base pairing confers a very unique property on the polynucleotide chains, which are said to be complementary to each other, so if the sequence of bases in one strand is known the sequence in the other can be predicted
- if each strand of a parental DNA acts as a template for the synthesis of a new strand, the two double stranded daughter DNA molecules produced would be identical to the parental molecule, and because of this the genetic implications of the structure became very clear
- the structure is made of two polynucleotide chains, where the backbone is constituted by sugar and phosphate and the bases project inside
- the two chains have anti-parallel polarity, meaning that if one chain has the polarity 5' to 3' the other has 3' to 5'

The base pairs

- the bases in the two strands are paired through hydrogen bonds forming base pairs (bp)
- adenine forms two hydrogen bonds with thymine from the opposite strand and vice versa, while guanine is bonded with cytosine with three hydrogen bonds
- as a result a purine always comes opposite a pyrimidine, which generates an approximately uniform distance between the two strands of the helix

Two chains running in opposite directions

Adenine paired with thymine by two hydrogen bonds

Guanine paired with cytosine by three hydrogen bonds

A purine always opposite a pyrimidine

Figure 5.B Double stranded polynucleotide chain

The two chains coiled in a right-handed fashion

Pitch of the helix, 3.4 nm

Roughly 10 base pairs in each turn

About 0.34 nm between one base pair and the next

Figure 5.C DNA double helix

The geometry of the helix

- the two chains are coiled in a right-handed fashion, the pitch of the helix being 3.4 nm, a nanometre being one billionth of a metre, and there are roughly 10 bp in each turn, so the distance between a bp in a helix is approximately 0.34 nm
- the plane of one base pair stacks over the other in the double helix, and this, in addition to the hydrogen bonds, confers stability on the helical structure
- comparing the structure of purines and pyrimidines shows why the distance between the two polynucleotide chains in DNA remains almost constant

5.4 The central dogma

How information flows

- the proposition of a double helix structure for DNA, and its simplicity in explaining the genetic implication, became revolutionary
- very soon Francis Crick proposed the central dogma in molecular biology
- it states that the genetic information flows from DNA to RNA to protein

Histone octamer, eight molecules of basic protein

Negatively charged DNA wrapped around it

About 200 bp of DNA helix in a typical nucleosome

Figure 5.D Nucleosome

Nucleosomes as the repeating unit of chromatin

Seen as beads on a string under the electron microscope

Thread-like stained bodies in the nucleus

Figure 5.E EM picture - 'Beads-on-String'

5.5 Packaging of the DNA helix

The problem of length

- in some viruses the flow of information is in the reverse direction, that is from RNA to DNA
- taking the distance between two consecutive base pairs as 0.34 nm, the length of the DNA double helix in a typical mammalian cell, found by multiplying the total number of bp by that distance, comes out to be approximately 2.2 metres, far greater than the dimension of a typical nucleus, which is about one millionth of a metre
- in prokaryotes such as E. coli, though they do not have a defined nucleus, the DNA is not scattered throughout the cell, but being negatively charged is held with some proteins that have positive charges in a region termed the nucleoid, where it is organised in large loops held by proteins

Histones and nucleosomes

- in eukaryotes this organisation is much more complex, with a set of positively charged basic proteins called histones, a protein acquiring charge depending on the abundance of amino acid residues with charged side chains
- histones are rich in the basic amino acid residues lysine and arginine, both of which carry positive charges in their side chains, and are organised to form a unit of eight molecules called a histone octamer
- the negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called a nucleosome, a typical nucleosome containing 200 bp of DNA helix
- nucleosomes constitute the repeating unit of a structure in the nucleus called chromatin, thread-like stained bodies seen in the nucleus, and the nucleosomes in chromatin are seen as a beads-on-string structure under the electron microscope

From chromatin to chromosome

- the beads-on-string structure in chromatin is packaged to form chromatin fibers, which are further coiled and condensed at the metaphase stage of cell division to form chromosomes
- the packaging of chromatin at a higher level requires an additional set of proteins, collectively referred to as non-histone chromosomal (NHC) proteins
- in a typical nucleus some regions of chromatin are loosely packed and stain light, and are referred to as euchromatin, while chromatin that is more densely packed and stains dark is called heterochromatin, euchromatin being transcriptionally active and heterochromatin inactive

5.6 The search for genetic material

Where the search had reached

- even though the discovery of nuclein by Meischer and the proposition of principles of inheritance by Mendel were almost at the same time, it took long to discover and prove that DNA acts as the genetic material
- by 1926 the quest to determine the mechanism for genetic inheritance had reached the molecular level, previous discoveries by Gregor Mendel, Walter Sutton, Thomas Hunt Morgan and numerous other scientists having narrowed the search to the chromosomes located in the nucleus of most cells
- but the question of what molecule was actually the genetic material had not been answered

Griffith's transforming principle

- in 1928 Frederick Griffith, in a series of experiments with Streptococcus pneumoniae, the bacterium responsible for pneumonia, witnessed a miraculous transformation in the bacteria, a living organism having changed in physical form during the course of his experiment
- when pneumococcus bacteria are grown on a culture plate some produce smooth shiny colonies (S) while others produce rough colonies (R), because the S strain bacteria have a mucous polysaccharide coat while the R strain does not, and mice infected with the virulent S strain die from pneumonia infection while mice infected with the R strain do not develop pneumonia
- Griffith was able to kill bacteria by heating them, and observed that heat-killed S strain bacteria injected into mice did not kill them

What Griffith concluded

- when he injected a mixture of heat-killed S and live R bacteria the mice died, and moreover he recovered living S bacteria from the dead mice
- he concluded that the R strain bacteria had somehow been transformed by the heat-killed S strain bacteria, some transforming principle transferred from the heat-killed S strain having enabled the R strain to synthesise a smooth polysaccharide coat and become virulent
- this must be due to the transfer of the genetic material, however the biochemical nature of that material was not defined from his experiments

5.7 The biochemical nature of the transforming principle

Avery, MacLeod and McCarty

- prior to their work, from 1933 to 1944, the genetic material was thought to be a protein, and they worked to determine the biochemical nature of the transforming principle in Griffith's experiment
- they purified biochemicals, that is proteins, DNA and RNA, from the heat-killed S cells to see which ones could transform live R cells into S cells, and discovered that DNA alone from S bacteria caused R bacteria to become transformed
- they also discovered that protein-digesting enzymes, the proteases, and RNA-digesting enzymes, the RNases, did not affect transformation, so the transforming substance was not a protein or RNA, while digestion with DNase did inhibit transformation, suggesting that DNA caused it, and they concluded that DNA is the hereditary material, though not all biologists were convinced

5.8 The genetic material is DNA

Hershey and Chase

- the unequivocal proof came from the experiments of Alfred Hershey and Martha Chase in 1952, who worked with viruses that infect bacteria, called bacteriophages
- a bacteriophage attaches to the bacterium and its genetic material then enters the bacterial cell, which treats the viral genetic material as if it were its own and subsequently manufactures more virus particles, and Hershey and Chase worked to discover whether it was protein or DNA from the viruses that entered the bacteria
- they grew some viruses on a medium containing radioactive phosphorus and others on a medium containing radioactive sulfur, so viruses grown with radioactive phosphorus contained radioactive DNA but not radioactive protein, because DNA contains phosphorus and protein does not, and viruses grown on radioactive sulfur contained radioactive protein but not radioactive DNA, because DNA does not contain sulfur

Phages grown with radioactive phosphorus, labelling their DNA

Phages grown with radioactive sulfur, labelling their protein

Viral coats removed by agitating the bacteria in a blender

Virus particles separated from bacteria in a centrifuge

Radioactivity found in the bacteria only from the DNA-labelled phages

Figure 5.F The Hershey-Chase experiment

What the experiment showed

- radioactive phages were allowed to attach to E. coli bacteria, and then, as the infection proceeded, the viral coats were removed from the bacteria by agitating them in a blender, the virus particles being separated from the bacteria by spinning them in a centrifuge
- bacteria infected with viruses that had radioactive DNA were radioactive, indicating that DNA was the material that passed from the virus to the bacteria
- bacteria infected with viruses that had radioactive proteins were not radioactive, indicating that proteins did not enter the bacteria from the viruses
- DNA is therefore the genetic material that is passed from virus to bacteria

5.9 Properties of the genetic material

Where RNA is the genetic material

- the debate between proteins and DNA as the genetic material was unequivocally resolved by the Hershey-Chase experiment, and it became an established fact that DNA acts as genetic material
- however it subsequently became clear that in some viruses RNA is the genetic material, for example the tobacco mosaic viruses and the QB bacteriophage
- the answer to why DNA is the predominant genetic material whereas RNA performs the dynamic functions of messenger and adapter has to be found from the differences between the chemical structures of the two nucleic acid molecules

5.10 The four criteria

What a molecule must do

- it should be able to generate its replica, that is replication
- it should be stable chemically and structurally
- it should provide the scope for slow changes, that is mutation, that are required for evolution
- it should be able to express itself in the form of Mendelian characters

Replication and stability

- because of the rule of base pairing and complementarity both the nucleic acids have the ability to direct their duplications, while other molecules in the living system such as proteins fail to fulfill the first criterion itself
- the genetic material should be stable enough not to change with different stages of life cycle, age or change in physiology, and stability was very evident in Griffith's transforming principle itself, where heat which killed the bacteria at least did not destroy some of the properties of the genetic material
- this can be explained in the light of DNA, in that the two strands, being complementary, if separated by heating come together again when appropriate conditions are provided
- further, the 2' OH group present at every nucleotide in RNA is a reactive group and makes RNA labile and easily degradable, and RNA is also now known to be catalytic and hence reactive, so DNA is chemically less reactive and structurally more stable than RNA, and among the two nucleic acids DNA is a better genetic material

Mutation and expression

- the presence of thymine in place of uracil also confers additional stability on DNA
- both DNA and RNA are able to mutate, and RNA, being unstable, mutates at a faster rate, so viruses having an RNA genome and a shorter life span mutate and evolve faster
- RNA can directly code for the synthesis of proteins and hence can easily express the characters, whereas DNA is dependent on RNA for the synthesis of proteins, the protein synthesising machinery having evolved around RNA

Parental double helix, its two strands separating

Each strand acting as a template for a new complementary strand

Two daughter molecules, each with one parental and one new strand

Hence semiconservative replication

Figure 5.G Watson-Crick model for semiconservative DNA replication

5.11 RNA world

Which came first

- both RNA and DNA can function as genetic material, but DNA, being more stable, is preferred for the storage of genetic information, while for the transmission of genetic information RNA is better
- RNA was the first genetic material, and there is now enough evidence to suggest that essential life processes such as metabolism, translation and splicing evolved around RNA
- RNA used to act as a genetic material as well as a catalyst, there being some important biochemical reactions in living systems that are catalysed by RNA catalysts and not by protein enzymes, but RNA being a catalyst was reactive and hence unstable
- therefore DNA has evolved from RNA with chemical modifications that make it more stable, and DNA, being double stranded and having a complementary strand, further resists changes by evolving a process of repair

5.12 Replication

The scheme Watson and Crick proposed

- while proposing the double helical structure they immediately proposed a scheme for replication, writing in 1953 that it had not escaped their notice that the specific pairing they had postulated immediately suggests a possible copying mechanism for the genetic material
- the scheme suggested that the two strands would separate and act as a template for the synthesis of new complementary strands
- after the completion of replication each DNA molecule would have one parental and one newly synthesised strand, and this scheme was termed semiconservative DNA replication

E. coli grown for many generations on heavy nitrogen

Cells transferred to a medium with normal nitrogen

Samples taken at intervals as the cells multiplied

DNA separated on caesium chloride density gradients

Bands of heavy, hybrid and light density read off

Figure 5.H Meselson and Stahl's Experiment

5.13 The experimental proof

The experiment of 1958

- it is now proven that DNA replicates semiconservatively, shown first in Escherichia coli and subsequently in higher organisms such as plants and human cells, Matthew Meselson and Franklin Stahl having performed the experiment in 1958
- they grew E. coli in a medium containing heavy nitrogen as the only nitrogen source for many generations, so that the heavy isotope was incorporated into newly synthesised DNA as well as other nitrogen containing compounds, and this heavy DNA molecule could be distinguished from normal DNA by centrifugation in a cesium chloride density gradient, the heavy isotope not being radioactive and separable only on the basis of density
- they then transferred the cells into a medium with normal nitrogen and took samples at various definite time intervals as the cells multiplied, extracting the DNA that remained as double-stranded helices, and the various samples were separated independently on caesium chloride gradients to measure the densities of the DNA
- recalling what centrifugal force is, and why a molecule with higher mass or density would sediment faster, is worth doing

What the gradients showed

- the DNA extracted from the culture one generation after the transfer, that is after 20 minutes, since E. coli divides in 20 minutes, had a hybrid or intermediate density
- DNA extracted after another generation, that is after 40 minutes, was composed of equal amounts of this hybrid DNA and of light DNA, which leaves the proportions of light and hybrid DNA after 80 minutes to be worked out
- very similar experiments involving the use of radioactive thymidine to detect the distribution of newly synthesised DNA in the chromosomes were performed on Vicia faba, the faba bean, by Taylor and colleagues in 1958, and proved that the DNA in chromosomes also replicates semiconservatively

5.14 The machinery and the enzymes

DNA polymerase

- in living cells such as E. coli the process of replication requires a set of catalysts, the main enzyme being referred to as DNA-dependent DNA polymerase, since it uses a DNA template to catalyse the polymerisation of deoxynucleotides
- these are highly efficient enzymes, as they have to catalyse polymerisation of a large number of nucleotides in a very short time, so that E. coli, which has only 4.6 million bp against the human diploid content of 6.6 billion bp, completes replication within 18 minutes, an average rate of approximately 2000 bp per second
- not only do these polymerases have to be fast, they also have to catalyse the reaction with a high degree of accuracy, since any mistake during replication would result in mutations
- energetically, replication is a very expensive process, and deoxyribonucleoside triphosphates serve dual purposes, acting as substrates and also providing energy for the polymerisation reaction, the two terminal phosphates being high-energy phosphates as in ATP

The replication fork

- in addition to DNA-dependent DNA polymerases, many additional enzymes are required to complete the process with a high degree of accuracy
- for long DNA molecules, since the two strands cannot be separated along their entire length because of the very high energy requirement, replication occurs within a small opening of the DNA helix, referred to as the replication fork
- the DNA-dependent DNA polymerases catalyse polymerisation only in one direction, that is 5’ to 3’, which creates additional complications at the replicating fork
- consequently, on the template with polarity 3’ to 5’ the replication is continuous, while on the template with polarity 5’ to 3’ it is discontinuous, and the discontinuously synthesised fragments are later joined by the enzyme DNA ligase

A small opening of the helix, the replication fork

Continuous synthesis on the 3' to 5' template

Discontinuous synthesis on the 5' to 3' template

Fragments later joined by DNA ligase

Figure 5.I Replicating Fork

5.15 Origin of replication

Where replication starts

- the DNA polymerases on their own cannot initiate the process of replication, and replication does not initiate randomly at any place in DNA
- there is a definite region in E. coli DNA where replication originates, and such regions are termed the origin of replication
- it is because of the requirement of the origin of replication that a piece of DNA, if it needs to be propagated during recombinant DNA procedures, requires a vector, the vectors providing the origin of replication
- in eukaryotes the replication of DNA takes place at the S-phase of the cell-cycle, and the replication of DNA and the cell division cycle should be highly coordinated, a failure in cell division after DNA replication resulting in polyploidy, a chromosomal anomaly

5.16 Transcription

What transcription is

- the process of copying genetic information from one strand of the DNA into RNA is termed transcription, and here also the principle of complementarity governs the process, except that adenosine now forms a base pair with uracil instead of thymine
- unlike replication, which once set in duplicates the total DNA of an organism, in transcription only a segment of DNA, and only one of the strands, is copied into RNA, which necessitates defining the boundaries that demarcate the region and the strand to be transcribed
- if both strands acted as a template they would code for RNA molecules with different sequences, since complementarity does not mean identical, so one segment of the DNA would code for two different proteins and complicate the genetic information transfer machinery
- second, the two RNA molecules, if produced simultaneously, would be complementary to each other and so form a double stranded RNA, which would prevent the RNA from being translated into protein and make the exercise of transcription a futile one

The three regions of a transcription unit

- a promoter
- the structural gene
- a terminator

Promoter at the 5' end, upstream of the structural gene

Structural gene between them

Terminator at the 3' end, downstream

Template strand and coding strand, of opposite polarity

Figure 5.J Schematic structure of a transcription unit

5.17 Template and coding strands

The convention

- since the two strands have opposite polarity and DNA-dependent RNA polymerase catalyses polymerisation in only one direction, that is 5' to 3', the strand that has the polarity 3' to 5' acts as a template and is referred to as the template strand
- the other strand, which has the polarity 5' to 3' and the sequence same as the RNA except for thymine in place of uracil, is displaced during transcription
- strangely, this strand, which does not code for anything, is referred to as the coding strand, all reference points in defining a transcription unit being made with it
- a hypothetical transcription unit runs 3' -ATGCATGCATGCATGCATGCATGC-5' on the template strand and 5' -TACGTACGTACGTACGTACGTACG-3' on the coding strand, which leaves the sequence of RNA transcribed from it to be written out

Promoter and terminator

- they flank the structural gene in a transcription unit
- the promoter is said to be located towards the 5' end, upstream of the structural gene, the reference being made with respect to the polarity of the coding strand, and it is a DNA sequence that provides a binding site for RNA polymerase, its presence also defining the template and coding strands, so that by switching its position with the terminator the definition of coding and template strands could be reversed
- the terminator is located towards the 3' end, downstream of the coding strand, and it usually defines the end of the process of transcription
- there are additional regulatory sequences that may be present further upstream or downstream to the promoter

RNA polymerase binding to the promoter, initiation

Polymerisation along the template, elongation

The terminator reached, the nascent RNA falling off

Initiation factor and termination factor associating transiently

Figure 5.K Process of Transcription in Bacteria

5.18 The transcription unit and the gene

What counts as a gene

- a gene is defined as the functional unit of inheritance, and though there is no ambiguity that genes are located on the DNA, it is difficult to literally define a gene in terms of DNA sequence, since the DNA sequence coding for a tRNA or rRNA molecule also defines a gene
- by defining a cistron as a segment of DNA coding for a polypeptide, the structural gene in a transcription unit could be said to be monocistronic, mostly in eukaryotes, or polycistronic, mostly in bacteria or prokaryotes
- in eukaryotes the monocistronic structural genes have interrupted coding sequences, that is the genes in eukaryotes are split, the coding or expressed sequences being defined as exons, those that appear in mature or processed RNA, interrupted by introns or intervening sequences, which do not appear in mature or processed RNA
- inheritance of a character is also affected by promoter and regulatory sequences of a structural gene, so sometimes the regulatory sequences are loosely defined as regulatory genes, even though they do not code for any RNA or protein

5.19 Types of RNA

Three RNAs in bacteria

- in bacteria there are three major types of RNA, mRNA or messenger RNA, tRNA or transfer RNA, and rRNA or ribosomal RNA, all three being needed to synthesise a protein in a cell
- the mRNA provides the template, tRNA brings amino acids and reads the genetic code, and rRNAs play structural and catalytic roles during translation
- there is a single DNA-dependent RNA polymerase that catalyses transcription of all types of RNA in bacteria, which binds to the promoter and initiates transcription, using nucleoside triphosphates as substrate

Primary transcript containing both exons and introns

Splicing, removing the introns and joining the exons

Capping, an unusual nucleotide added at the 5' end

Tailing, adenylate residues added at the 3' end

Fully processed mRNA transported out of the nucleus

Figure 5.L Process of Transcription in Eukaryotes

5.20 Initiation, elongation and termination

The two factors

- the polymerase polymerises in a template dependent fashion following the rule of complementarity, and it also facilitates opening of the helix and continues elongation, only a short stretch of RNA remaining bound to the enzyme, and once the polymerase reaches the terminator region the nascent RNA falls off, and so does the RNA polymerase, which results in termination
- the RNA polymerase is only capable of catalysing the process of elongation, which raises the question of how it catalyses all three steps
- it associates transiently with an initiation-factor and a termination-factor to initiate and terminate transcription respectively, association with these factors altering the specificity of the RNA polymerase
- in bacteria, since the mRNA does not require any processing to become active, and since transcription and translation take place in the same compartment, there being no separation of cytosol and nucleus, many times translation can begin much before the mRNA is fully transcribed, so the two can be coupled in bacteria

5.21 Two complexities in eukaryotes

Polymerases, and split genes

- there are at least three RNA polymerases in the nucleus, in addition to the RNA polymerase found in the organelles, with a clear cut division of labour
- the second complexity is that the primary transcripts contain both the exons and the introns and are non-functional

5.22 Processing the transcript

The three polymerases

- RNA polymerase I transcribes rRNAs, that is 28S, 18S and 5.8S
- RNA polymerase III is responsible for the transcription of tRNA, 5srRNA and snRNAs, the small nuclear RNAs
- RNA polymerase II transcribes the precursor of mRNA, the heterogeneous nuclear RNA (hnRNA)

Splicing, capping and tailing

- the primary transcript is subjected to a process called splicing, where the introns are removed and the exons joined in a defined order
- hnRNA undergoes additional processing called capping and tailing, in capping an unusual nucleotide, methyl guanosine triphosphate, being added to the 5' end, and in tailing adenylate residues, 200 to 300 of them, being added at the 3' end in a template independent manner
- it is the fully processed hnRNA, now called mRNA, that is transported out of the nucleus for translation
- the split-gene arrangement probably represents an ancient feature of the genome, the presence of introns is reminiscent of antiquity, and the process of splicing represents the dominance of the RNA-world

5.23 The genetic code

Why a code was needed, and Gamow's argument

- during replication and transcription a nucleic acid was copied to form another nucleic acid, so these processes are easy to conceptualise on the basis of complementarity, whereas translation requires transfer of genetic information from a polymer of nucleotides to synthesise a polymer of amino acids
- no complementarity exists between nucleotides and amino acids, nor could any be drawn theoretically, yet there was ample evidence that changes in nucleic acids were responsible for changes in amino acids in proteins, which led to the proposition of a genetic code
- the proposition and deciphering of the code were most challenging, and required the involvement of physicists, organic chemists, biochemists and geneticists
- it was George Gamow, a physicist, who argued that since there are only 4 bases, and if they have to code for 20 amino acids, the code should constitute a combination of bases, and should be made up of three nucleotides, a bold proposition because a permutation combination of 4 bases taken three at a time would generate 64 codons, many more than required

A checker-board of all 64 codons

61 codons coding for amino acids

3 codons coding for none, and acting as stop codons

Several amino acids coded by more than one codon

Table 5.A The Codons for the Various Amino Acids

5.24 Deciphering the code

Who cracked it

- providing proof that the codon was a triplet was a more daunting task
- the chemical method developed by Har Gobind Khorana was instrumental in synthesising RNA molecules with defined combinations of bases, that is homopolymers and copolymers
- Marshall Nirenberg's cell-free system for protein synthesis finally helped the code to be deciphered, and Severo Ochoa's enzyme, polynucleotide phosphorylase, was also helpful in polymerising RNA with defined sequences in a template independent manner

The salient features

- the codon is a triplet, 61 codons coding for amino acids and 3 codons coding for none, hence functioning as stop codons
- some amino acids are coded by more than one codon, hence the code is degenerate, and the codon is read in mRNA in a contiguous fashion, there being no punctuations
- the code is nearly universal, so that from bacteria to human UUU would code for phenylalanine, some exceptions having been found in mitochondrial codons and in some protozoans
- AUG has dual functions, coding for methionine and also acting as initiator codon, while UAA, UAG and UGA are stop terminator codons

Read the code both ways

- given the sequence of nucleotides AUG UUU UUC UUC UUU UUU UUC in an mRNA, predict the sequence of amino acids coded by it, taking help of the checkerboard
- now try the opposite: given the sequence of amino acids Met-Phe-Phe-Phe-Phe-Phe-Phe coded by an mRNA, predict the nucleotide sequence in the RNA
- whether there is any difficulty in predicting the opposite, and which two properties of the genetic code that difficulty illustrates, are worth working out

5.25 Mutations and the genetic code

Point mutations

- the relationships between genes and DNA are best understood by mutation studies, and the effects of large deletions and rearrangements in a segment of DNA are easy to comprehend, since they may result in loss or gain of a gene and so of a function
- a classical example of point mutation is a change of a single base pair in the gene for the beta globin chain that results in the change of the amino acid residue glutamate to valine
- it results in a diseased condition called sickle cell anemia

Reading frames

- consider a statement made up of words each having three letters, like the genetic code: RAM HAS RED CAP
- inserting a letter B between HAS and RED and rearranging gives RAM HAS BRE DCA P, inserting two letters gives RAM HAS BIR EDC AP, and inserting three letters together gives RAM HAS BIG RED CAP
- the same exercise repeated by deleting the letters R, E and D one by one and rearranging gives RAM HAS EDC AP, then RAM HAS DCA P, then RAM HAS CAP

Anticodon loop, its bases complementary to the code

Amino acid acceptor end, binding a specific amino acid

A clover-leaf shape in the secondary structure

An inverted L in the actual compact molecule

Figure 5.M tRNA - the adapter molecule

What the exercise shows

- insertion or deletion of one or two bases changes the reading frame from the point of insertion or deletion, and such mutations are referred to as frameshift insertion or deletion mutations
- insertion or deletion of three or a multiple of three bases inserts or deletes one or multiple codons, hence one or multiple amino acids, and the reading frame remains unaltered from that point onwards

5.26 tRNA, the adapter molecule

Crick's postulate

- from the very beginning of the proposition of the code it was clear to Francis Crick that there has to be a mechanism to read the code and also to link it to the amino acids, because amino acids have no structural specialities to read the code uniquely, so he postulated the presence of an adapter molecule that would on one hand read the code and on the other bind to specific amino acids
- the tRNA, then called sRNA or soluble RNA, was known before the genetic code was postulated, but its role as an adapter molecule was assigned much later
- tRNA has an anticodon loop that has bases complementary to the code, and an amino acid acceptor end to which it binds amino acids, tRNAs being specific for each amino acid, while for initiation there is another specific tRNA referred to as the initiator tRNA, and there are no tRNAs for stop codons

5.27 Translation

Charging the tRNA

- translation refers to the process of polymerisation of amino acids to form a polypeptide, the order and sequence of amino acids being defined by the sequence of bases in the mRNA
- the amino acids are joined by a bond known as a peptide bond, and the formation of a peptide bond requires energy, so in the first phase itself amino acids are activated in the presence of ATP and linked to their cognate tRNA, a process commonly called charging of tRNA, or aminoacylation of tRNA to be more specific

mRNA held between the two ribosomal subunits

Charged tRNAs bringing amino acids to the codons

Two sites in the large subunit, holding successive amino acids

Peptide bond formed, the polypeptide growing

Figure 5.N Translation

5.28 The ribosome and the translational unit

The cellular factory

- if two charged tRNAs are brought close enough the formation of a peptide bond between them would be favoured energetically, and the presence of a catalyst would enhance the rate of peptide bond formation
- the cellular factory responsible for synthesising proteins is the ribosome, which consists of structural RNAs and about 80 different proteins, and in its inactive state exists as two subunits, a large and a small one, translation beginning when the small subunit encounters an mRNA
- there are two sites in the large subunit for subsequent amino acids to bind to and thus be close enough for the formation of a peptide bond, and the ribosome also acts as a catalyst for that formation, the 23S rRNA in bacteria being the enzyme, a ribozyme

The translational unit, and the three phases

- a translational unit in mRNA is the sequence of RNA flanked by the start codon (AUG) and the stop codon, which codes for a polypeptide, while an mRNA also has additional sequences that are not translated, referred to as untranslated regions (UTR), present at both the 5' end before the start codon and the 3' end after the stop codon, and required for efficient translation
- for initiation the ribosome binds to the mRNA at the start codon, which is recognised only by the initiator tRNA
- during elongation, complexes composed of an amino acid linked to tRNA sequentially bind to the appropriate codon in mRNA by forming complementary base pairs with the tRNA anticodon, the ribosome moving from codon to codon along the mRNA and amino acids being added one by one
- at the end a release factor binds to the stop codon, terminating translation and releasing the complete polypeptide from the ribosome

5.29 Regulation of gene expression

Where regulation can act in eukaryotes

- at the transcriptional level, that is the formation of the primary transcript
- at the processing level, that is the regulation of splicing
- at the transport of mRNA from nucleus to the cytoplasm
- and at the translational level

5.30 What decides whether a gene is expressed

Conditions, not caprice

- the genes in a cell are expressed to perform a particular function or set of functions, so if the enzyme beta-galactosidase is synthesised by E. coli it is used to catalyse the hydrolysis of the disaccharide lactose into galactose and glucose, which the bacteria use as a source of energy, and if the bacteria do not have lactose around them they would no longer require the synthesis of that enzyme
- so in simple terms it is the metabolic, physiological or environmental conditions that regulate the expression of genes, and the development and differentiation of embryo into adult organisms are also a result of the coordinated regulation of expression of several sets of genes

Operators and repressors

- in prokaryotes, control of the rate of transcriptional initiation is the predominant site for control of gene expression, and in a transcription unit the activity of RNA polymerase at a given promoter is regulated by interaction with accessory proteins, which affect its ability to recognise start sites, and these regulatory proteins can act both positively, as activators, and negatively, as repressors
- the accessibility of promoter regions of prokaryotic DNA is in many cases regulated by the interaction of proteins with sequences termed operators, the operator region being adjacent to the promoter elements in most operons
- in most cases the sequences of the operator bind a repressor protein, each operon having its specific operator and specific repressor, so the lac operator is present only in the lac operon and interacts specifically with the lac repressor only

5.31 The lac operon

The genes it contains

- the elucidation of the lac operon was a result of a close association between a geneticist, Francois Jacob, and a biochemist, Jacque Monod, who were the first to elucidate a transcriptionally regulated system
- in the lac operon a polycistronic structural gene is regulated by a common promoter and regulatory genes, an arrangement very common in bacteria and referred to as an operon, other examples being the trp, ara, his and val operons
- the operon consists of one regulatory gene, the i gene, where the term i does not refer to inducer but is derived from the word inhibitor, and three structural genes z, y and a
- the i gene codes for the repressor, the z gene for beta-galactosidase, primarily responsible for the hydrolysis of lactose into galactose and glucose, the y gene for permease, which increases permeability of the cell to β-galactosides, and the a gene encodes a transacetylase, so all three gene products are required for metabolism of lactose

The i gene, coding for the repressor

Promoter and operator ahead of the structural genes

z, y and a, the three structural genes

Repressor bound to the operator, blocking RNA polymerase

Inducer inactivating the repressor, transcription proceeding

Figure 5.0 The lac Operon

5.32 Induction of the operon

Lactose as inducer

- lactose is the substrate for the enzyme beta-galactosidase and it regulates switching the operon on and off, hence it is termed the inducer
- in the absence of a preferred carbon source such as glucose, if lactose is provided in the growth medium it is transported into the cells through the action of permease, a very low level of expression of the lac operon having to be present in the cell all the time, otherwise lactose cannot enter the cells
- the repressor of the operon is synthesised constitutively from the i gene, binds to the operator region and prevents RNA polymerase from transcribing the operon
- in the presence of an inducer such as lactose or allolactose the repressor is inactivated by interaction with the inducer, which allows RNA polymerase access to the promoter and transcription proceeds

Positive and negative

- essentially, regulation of the lac operon can also be visualised as regulation of enzyme synthesis by its substrate
- glucose or galactose cannot act as inducers for the lac operon, which leaves how long the operon would be expressed in the presence of lactose to be worked out
- regulation of the lac operon by repressor is referred to as negative regulation, and the operon is under the control of positive regulation as well

5.33 Human Genome Project

Why it was attempted

- it is the sequence of bases in DNA that determines the genetic information of a given organism, so the genetic make-up of an organism or an individual lies in the DNA sequences, and if two individuals differ their DNA sequences should also differ, at least at some places
- these assumptions led to the quest of finding out the complete DNA sequence of the human genome, and with the establishment of genetic engineering techniques, where it was possible to isolate and clone any piece of DNA, and the availability of simple and fast techniques for determining DNA sequences, a very ambitious project was launched in 1990
- the Human Genome Project (HGP) was called a mega project, since the human genome is said to have approximately 3 billion bp, and at an estimated cost of US $ 3 per bp the total cost would be approximately 9 billion US dollars
- further, if the obtained sequences were stored in typed form in books of 1000 letters a page and 1000 pages a book, then 3300 such books would be required to store the information from a single human cell, and the enormous amount of data also necessitated high speed computational devices for data storage, retrieval and analysis, HGP being closely associated with the rapid development of a new area in biology called bioinformatics

The goals

- identify all the approximately 20,000 to 25,000 genes in human DNA
- determine the sequences of the 3 billion chemical base pairs that make up human DNA, store this information in databases, and improve tools for data analysis
- transfer related technologies to other sectors, such as industries
- address the ethical, legal and social issues (ELSI) that may arise from the project

Total DNA isolated and broken into random fragments

Fragments cloned in bacteria or yeast, using BAC and YAC vectors

Each fragment sequenced on an automated DNA sequencer

Overlapping regions used to align the sequences

Computer programs assembling the whole

Figure 5.P A representative diagram of human genome project

5.34 How the project was run

Who did it

- the Human Genome Project was a 13-year project coordinated by the U.S. Department of Energy and the National Institute of Health, and during its early years the Wellcome Trust of the U.K. became a major partner, with additional contributions from Japan, France, Germany, China and others, the project being completed in 2003
- knowledge about the effects of DNA variations among individuals can lead to revolutionary new ways to diagnose, treat and someday prevent the thousands of disorders that affect human beings
- besides providing clues to understanding human biology, learning about non-human organisms' DNA sequences can lead to an understanding of their natural capabilities that can be applied towards solving challenges in health care, agriculture, energy production and environmental remediation, and many non-human model organisms such as bacteria, yeast, Caenorhabditis elegans, Drosophila and plants such as rice and Arabidopsis have also been sequenced

The two approaches

- one approach focused on identifying all the genes that are expressed as RNA, referred to as expressed sequence tags (ESTs)
- the other took the blind approach of simply sequencing the whole set of genome that contained all the coding and non-coding sequence, and later assigning different regions in the sequence with functions, a term referred to as sequence annotation
- for sequencing, the total DNA from a cell is isolated and converted into random fragments of relatively smaller sizes and cloned in a suitable host using specialised vectors, the cloning resulting in amplification of each piece so that it could subsequently be sequenced with ease, the commonly used hosts being bacteria and yeast and the vectors called BAC, bacterial artificial chromosomes, and YAC, yeast artificial chromosomes

Assembling and mapping

- another challenging task was assigning the genetic and physical maps on the genome, generated using information on polymorphism of restriction endonuclease recognition sites and some repetitive DNA sequences known as microsatellites

5.35 Salient features of the human genome

The numbers

- the human genome contains 3164.7 million bp, and the average gene consists of 3000 bases, though sizes vary greatly, the largest known human gene being dystrophin at 2.4 million bases
- the total number of genes is estimated at 30,000, much lower than previous estimates of 80,000 to 1,40,000 genes, and almost all, 99.9 per cent, of nucleotide bases are exactly the same in all humans
- the functions are unknown for over 50 per cent of the discovered genes, and less than 2 per cent of the genome codes for proteins
- chromosome 1 has the most genes, 2968, and the Y has the fewest, 231

Repeats and snips

- repeated sequences make up a very large portion of the human genome
- repetitive sequences are stretches of DNA sequences that are repeated many times, sometimes hundreds to thousands of times, and are thought to have no direct coding functions, but they shed light on chromosome structure, dynamics and evolution
- scientists have identified about 1.4 million locations where single-base DNA differences, that is single nucleotide polymorphisms or SNPs, pronounced snips, occur in humans, and this information promises to revolutionise the processes of finding chromosomal locations for disease-associated sequences and tracing human history

5.36 Applications and future challenges

The scale of the task

- deriving meaningful knowledge from the DNA sequences will define research through the coming decades, leading to our understanding of biological systems
- this enormous task will require the expertise and creativity of tens of thousands of scientists from varied disciplines in both the public and private sectors worldwide
- one of the greatest impacts of having the human genome sequence may well be enabling a radically new approach to biological research, since in the past researchers studied one or a few genes at a time

A broader scale

- with whole-genome sequences and new high-throughput technologies we can approach questions systematically and on a much broader scale
- we can study all the genes in a genome, for example all the transcripts in a particular tissue or organ or tumor
- or how tens of thousands of genes and proteins work together in interconnected networks to orchestrate the chemistry of life

5.37 DNA fingerprinting

Why it is needed

- 99.9 per cent of base sequence among humans is the same, and it is the differences in sequence of DNA which make every individual unique in their phenotypic appearance
- if one aims to find out genetic differences between two individuals or among individuals of a population, sequencing the DNA every time would be a daunting and expensive task, so DNA fingerprinting is a very quick way to compare the DNA sequences of any two individuals
- it involves identifying differences in some specific regions in DNA sequence called repetitive DNA, because in these sequences a small stretch of DNA is repeated many times, and these repetitive DNA are separated from bulk genomic DNA as different peaks during density gradient centrifugation, the bulk DNA forming a major peak and the other small peaks being referred to as satellite DNA

Satellites, and why any tissue will do

- depending on base composition, length of segment and number of repetitive units, the satellite DNA is classified into many categories such as micro-satellites and mini-satellites, and these sequences normally do not code for any proteins but form a large portion of the human genome, showing a high degree of polymorphism and forming the basis of DNA fingerprinting
- since DNA from every tissue, such as blood, hair-follicle, skin, bone, saliva and sperm, from an individual shows the same degree of polymorphism, they become a very useful identification tool in forensic applications
- further, as the polymorphisms are inheritable from parents to children, DNA fingerprinting is the basis of paternity testing in case of disputes

5.38 DNA polymorphism

Where variation comes from

- as polymorphism in DNA sequence is the basis of genetic mapping of the human genome as well as of DNA fingerprinting, it is essential to understand what DNA polymorphism means, and polymorphism, that is variation at genetic level, arises due to mutations
- new mutations may arise in an individual either in somatic cells or in the germ cells, and if a germ cell mutation does not seriously impair an individual's ability to have offspring who can transmit the mutation, it can spread to the other members of the population through sexual reproduction
- allelic sequence variation has traditionally been described as a DNA polymorphism if more than one variant at a locus occurs in the human population with a frequency greater than 0.01, so in simple terms, if an inheritable mutation is observed in a population at high frequency, it is referred to as DNA polymorphism
- the probability of such variation being observed in non-coding DNA sequence would be higher, as mutations there may not have any immediate effect on an individual's reproductive ability, and these mutations keep accumulating generation after generation, forming one of the bases of variability, while there is a variety of different types of polymorphism ranging from single nucleotide change to very large scale changes

5.39 The technique

Alec Jeffreys and the VNTR

- the technique was initially developed by Alec Jeffreys, who used a satellite DNA as probe that shows a very high degree of polymorphism, called Variable Number of Tandem Repeats (VNTR)
- the technique as used earlier involved Southern blot hybridisation using radiolabelled VNTR as a probe
- it included isolation of DNA, digestion of DNA by restriction endonucleases, separation of DNA fragments by electrophoresis, transferring or blotting of separated DNA fragments to synthetic membranes such as nitrocellulose or nylon, hybridisation using labelled VNTR probe, and detection of hybridised DNA fragments by autoradiography

Chromosomes carrying different copy numbers of VNTR

The two alleles of a chromosome differing in copy number

Bands of differing size on the autoradiogram

The pattern from the crime scene matching individual B, not A

Figure 5.Q Schematic representation of DNA fingerprinting : Few representative chromosomes have been shown to contain different copy number of VNTR. For the sake of understanding different colour schemes have been used to trace the origin of each band in the gel. The two alleles (paternal and maternal) of a chromosome also contain different copy numbers of VNTR. It is clear that the banding pattern of DNA from crime scene matches with individual B, and not with A.

5.40 The VNTR and its bands

What a mini-satellite is

- the VNTR belongs to a class of satellite DNA referred to as mini-satellite, a small DNA sequence arranged tandemly in many copy numbers
- the copy number varies from chromosome to chromosome in an individual, and the numbers of repeat show a very high degree of polymorphism, so the size of a VNTR varies from 0.1 to 20 kb
- consequently, after hybridisation with the VNTR probe, the autoradiogram gives many bands of differing sizes, which give a characteristic pattern for an individual DNA, differing from individual to individual in a population except in the case of monozygotic identical twins

How far it now reaches

- the sensitivity of the technique has been increased by the use of polymerase chain reaction, so that DNA from a single cell is enough to perform DNA fingerprinting analysis
- in addition to application in forensic science it has much wider application, such as in determining population and genetic diversities
- currently, many different probes are used to generate DNA fingerprints

5.41 Summary

DNA, RNA and replication

- nucleic acids are long polymers of nucleotides, and while DNA stores genetic information RNA mostly helps in transfer and expression of information, and though both function as genetic material, DNA, being chemically and structurally more stable, is a better genetic material, although RNA is the first to evolve and DNA was derived from RNA
- the hallmark of the double stranded helical structure of DNA is the hydrogen bonding between the bases from opposite strands, the rule being that adenine pairs with thymine through two hydrogen bonds and guanine with cytosine through three, which makes one strand complementary to the other
- the DNA replicates semiconservatively, the process being guided by the complementary hydrogen bonding, and a segment of DNA that codes for RNA may in simplistic terms be referred to as a gene

Transcription and translation

- during transcription one of the strands of DNA acts as a template to direct the synthesis of complementary RNA, and in bacteria the transcribed mRNA is functional and can be directly translated, while in eukaryotes the gene is split, the coding sequences or exons being interrupted by non-coding sequences or introns, which are removed while the exons are joined to produce functional RNA by splicing
- the messenger RNA contains the base sequences that are read in a combination of three to code for an amino acid, and the genetic code is read again on the principle of complementarity by tRNA, which acts as an adapter molecule, there being specific tRNAs for every amino acid, binding to a specific amino acid at one end and pairing through hydrogen bonding with codes on mRNA through their anticodons
- the site of translation is the ribosome, which binds to mRNA and provides a platform for the joining of amino acids, one of the rRNAs acting as a catalyst for peptide bond formation, an example of an RNA enzyme or ribozyme, and translation is a process that has evolved around RNA, indicating that life began around RNA

Regulation, the genome and fingerprinting

- since transcription and translation are energetically very expensive processes they have to be tightly regulated, and regulation of transcription is the primary step for regulation of gene expression, while in bacteria more than one gene is arranged together and regulated in units called operons, the lac operon being the prototype, coding for genes responsible for metabolism of lactose and regulated by the amount of lactose in the medium, so that this regulation can also be viewed as regulation of enzyme synthesis by its substrate
- the Human Genome Project was a mega project that aimed to sequence every base in the human genome, which has yielded much new information and opened many new areas and avenues
- DNA fingerprinting is a technique to find out variations in individuals of a population at DNA level, working on the principle of polymorphism in DNA sequences, with immense applications in the field of forensic science, genetic biodiversity and evolutionary biology

5.42 Exercises

From this chapter

1. Group the following as nitrogenous bases and nucleosides: Adenine, Cytidine, Thymine, Guanosine, Uracil and Cytosine.
2. If a double stranded DNA has 20 per cent of cytosine, calculate the per cent of adenine in the DNA.
3. If the sequence of one strand of DNA is written as follows: 5' -ATGCATGCATGCATGCATGCATGCATGC-3' Write down the sequence of complementary strand in 5' →3' direction.
4. If the sequence of the coding strand in a transcription unit is written as follows: 5' -ATGCATGCATGCATGCATGCATGCATGC-3' Write down the sequence of mRNA.
5. Which property of DNA double helix led Watson and Crick to hypothesise semi-conservative mode of DNA replication? Explain.
6. Depending upon the chemical nature of the template (DNA or RNA) and the nature of nucleic acids synthesised from it (DNA or RNA), list the types of nucleic acid polymerases.
7. How did Hershey and Chase differentiate between DNA and protein in their experiment while proving that DNA is the genetic material?
8. Differentiate between the followings: (a) Repetitive DNA and Satellite DNA (b) mRNA and tRNA (c) Template strand and Coding strand
9. List two essential roles of ribosome during translation.
10. In the medium where E. coli was growing, lactose was added, which induced the lac operon. Then, why does lac operon shut down some time after addition of lactose in the medium?
11. Explain (in one or two lines) the function of the followings: (a) Promoter (b) tRNA (c) Exons
12. Why is the Human Genome project called a mega project?
13. What is DNA fingerprinting? Mention its application.
14. Briefly describe the following: (a) Transcription (b) Polymorphism (c) Translation (d) Bioinformatics